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Gene expression

A Bayesian framework for the inference of gene regulatory networks from time and pseudo-time series data

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Abstract

Motivation: Molecular profiling techniques have evolved to single-cell assays, where dense molecular profiles are screened simultaneously for each cell in a population. High-throughput single-cell experiments from a heterogeneous population of cells can be experimentally and computationally sorted as a sequence of samples pseudo-temporally ordered samples. The analysis of these datasets, comprising a large number of samples, has the potential to uncover the dynamics of the underlying regulatory programmes.

Results: We present a novel approach for modelling and inferring gene regulatory networks from high-throughput time series and pseudo-temporally sorted single-cell data. Our method is based on a first-order autoregressive moving-average model and it infers the gene regulatory network within a variational Bayesian framework. We validate our method with synthetic data and we apply it to single cell qPCR and RNA-Seq data for mouse embryonic cells and hematopoietic cells in zebra fish.

Availability and implementation: The method presented in this article is available at https://github.com/mscastillo/GRNVBEM.

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1 Introduction

Gene expression is a complex process whereby proteins are synthesized to control cell functions. During the earliest stage of gene expression, the biological information encoded in the DNA is transcribed into RNA. Subsequently, the RNA is processed and translated into a functional protein that regulates the metabolism of the cell (Shapiro *et al.*, 2009). Molecular profiling techniques have evolved to high-throughput single-cell (HTSC), where gene expression is quantified for a large population of cells. Traditional methods for transcriptome quantification, such as quantitative polimerase chain reaction (qPCR), are highly sensitive and preferred for interrogating a relatively small number of genes in single-cell assays (Wong *et al.*, 2005). In contrast, microarrays and RNA-Seq offer a higher multi-plex platform with a wider genome coverage (Wilson *et al.*, 2015).

One of the aims of Systems Biology is to understand how gene regulatory mechanisms work through the analysis of high-throughput data. One model that tries to explain how genes and their products interact at different levels is the gene regulatory network (GRN) (Das *et al.*, 2008). A GRN is a graphical model where genes are represented as nodes and their regulatory relationships as edges. GRN reverse engineering approaches attempt to infer GRNs from high-throughput data using statistical and computational methods. A plethora of GRN inference methods have been proposed during the last decade (Marbach *et al.*, 2012) and they can be divided into two main groups: static and the dynamic methods

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(Gardner *et al.*, 2005). A static method infers causal relationships between genes considering a steady-state, whilst dynamic methods take the time dependency into account. Dynamic models are considered more robust than the static ones, since they consider the temporal dimension of the problem. However, the inclusion of time as a variable makes dynamic approaches more sensitive to noise than the static ones (Lopes *et al.*, 2012).

Despite the fact that HTSC data does not constitute a time series, recent computational approaches have been successfully applied to dissect different subpopulations within the cell culture (Buettner *et al.*, 2012; Wilson *et al.*, 2015) and to capture the hierarchical organization within the cell population (Wilson *et al.*, 2015). These types of analyses are able to uncover the temporal dimension of the data and sort HTSC experiments according to a biologically meaningful pseudo-temporal index. HTSC sorting techniques coupled with dynamic GRN inference methods combine both the robustness of time series analyses and the high availability of data within single-cell experiments.

In this study, we address the problem of modelling and inferring GRN from time-series data and we extend it to the analysis of pseudo-time series data from HTSC experiments. We review the method proposed in Tienda-Luna *et al.* (2007) and we present a novel mathematical approach that fits the noisy nature of expression data better. Specifically, we consider a first-order autoregressive moving-average (AR1MA1) model to fit gene expression data and we propose a variational-Bayesian framework for the inference of the GRN. We have applied the proposed method to synthetic data and we have compared its performance with other approaches in the context of information retrieval. Additionally, we have applied the new method to the inference of GRN on early embryogenesis in mice using pseudo-temporally sorted qPCR data and RNA-Seq data for zebra fish as they undergo thrombocyte lineage commitment (Macaulay *et al.*, 2016). The resulting GRNs successfully captures known regulatory mechanisms described in the literature and depicts novel interactions which have not so far been described.

2 Materials and methods

Gene expression data are commonly expressed as $\log_2$-ratios relative to a reference level, namely fold change. The reference level could be a wild-type sample against an experimental condition it is compared with. On the other hand, the experimental sample could be generated by culturing cells under different conditions, phenotypes or drug treatments. Even wild-type profiles can be transformed to $\log_2$-ratios without any experimental sample, by centralizing data against the average expression in the whole experiment or normalizing according to housekeeping genes (Schmittgen *et al.*, 2008).

One of the advantages of using fold changes rather than raw intensities is that they provide a comprehensive symmetric measurement, where the absolute value denotes magnitude and the sign the direction of the change. Additionally, all the methods modelling this type of data are extensible to other types of molecular profiles where fold changes can be computed (Ritchie *et al.*, 2015).

2.1 The GRN model

GRNs have been used in Systems Biology to describe gene expression from a phenomenological point of view (Das *et al.*, 2008). In a GRN, the activation or inhibition of a gene, referred to as a child, depends on the expression of others presented in the network, its parents. This structure is conveniently modelled by a Bayesian network, a graphical model that represents probabilistic relationships (the edges of the network) amongst the genes (the nodes) (Hartemink *et al.*, 2005).

Given a set of G genes, the GRN can be fully characterized by the next two features (Ribeiro *et al.*, 2006). First, the network topology that represents the linkage pattern of the network. This logical structure has been formally described by a set of binary variables as

$$\mathbf{x}_i = [x_i(1), \ldots, x_i(G)]^\top \in \{0,1\}^{G\times 1}, \forall i \tag{1}$$

where $x_i(j) = 1$ specifies that the j-th gene is a parent of the th one; and $x_i(j) = 0$ otherwise.

Second, the regulatory effects that specifies the strength and type of interaction (activation or inhibition) amongst the genes. This scheme has been mathematically represented by an additional set of variables, referred to as weights, that are denoted as

$$\boldsymbol{\omega}_i = [\omega_i(1), \ldots, \omega_i(G)]^\top \in R^{G\times 1}, \forall i \tag{2}$$

where $\omega_i(j) > 0$ denotes activation and $\omega_i(j) < 0$ inhibition.

2.2 The observational model

Let $\mathbf{Y}$ be a dataset with G genes and $N + 1$ samples expressed in a matrix form where $y_i(n)$ is the fold change of the th gene at the nth time point. Autoregressive models have been previously considered for modelling microarray time-series (Morrissey *et al.*, 2010). Specifically, in (Tienda-Luna *et al.*, 2007) we considered a first-order autoregressive (AR1) model, where the fold change of the th gene at the nth time is a linear combination of the observations of its parents at the immediate previous time.

Fold changes are observations of the actual changes in expression affected by experimental noise (Raser *et al.*, 2005) as well as other sources of errors typical of the stochastic nature of the biological processes (Newman *et al.*, 2006). The AR1 model establishes relationships amongst data and it assumes that the noise constitutes a part of the actual regulatory process, underestimating the noise and leading to conclusions that may be incorrect.

In this article, we propose an approach that fits data by taking into account the difference between the noisy observations and the actual changes in expression levels. Specifically, we propose a linear model that represents the observations as a linear combination of the actual fold change, whose value is unknown and will be denoted by the variable $z_i(n)$, at the immediate previous time sample as

$$y_i(n) = \sum_{j=1}^{G} z_j(n-1)\omega_i(j)x_i(j) + e_i(n) \tag{3}$$

with $e_i(n)$ being the error at the current time sample, which is modelled as an independent and identically distributed Gaussian with mean zero and unknown variance as

$$p(e_i(n)) = \mathcal{N}\left(e_i(n) \mid 0, \sigma_i^2\right), \forall n. \tag{4}$$

The actual fold change $z_i(n)$ constitutes another unknown variable that can be expressed in terms of the data and the noise as $z_i(n) = y_i(n) - e_i(n)$, allowing us to rewrite the observational model in (3) as a AR1MA1 model as

$$y_i(n) = \sum_{j=1}^{G} \left(y_j(n-1) - e_j(n-1)\right)\omega_i(j)x_i(j) + e_i(n) \tag{5}$$

3 Reverse engineering GRN

Reverse engineering is the process of elucidating the dynamic of any system by reasoning backwards from its observation (Hartemink

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et al., 2005). In reverse engineering GRN, the structure of the network is inferred from molecular profiling data. Depending on the experimental design, inferences based on single-cell experiments will allow the exploration of the dynamics of the cell differentiation processes, the response to an external stimulus and reveal the hierarchical structure of the transcriptional programme.

3.1 Bayesian framework

Within the Bayesian framework, binary variables $\mathbf{x}_i$ describing the topology of the network are considered as hidden variables that describe the observation of the -th gene. Although the weights $\boldsymbol{\omega}_i$ are also hidden variables, they will be treated as parameters for mathematical convenience. Therefore, $\boldsymbol{\theta}_i = \{\boldsymbol{\omega}_i, \sigma_i^2\}$ denotes the set of parameters of the AR1MA1 model presented above.

The Bayesian approach to reverse GRN engineering relies on the optimization of the posterior $p(\mathbf{x}_i, \boldsymbol{\theta}_i \,|\, \mathbf{Y})$ to estimate the most probable value *a posterori* (MAP) of the hidden variables and the model parameters as

$$\left\{\widehat{\mathbf{x}}_{i,\mathrm{MAP}}, \widehat{\theta}_{i,\mathrm{MAP}}\right\} = \arg\max_{\mathbf{x}_i, \boldsymbol{\theta}_i} p(\mathbf{x}_i, \boldsymbol{\theta}_i \,|\, \mathbf{Y}). \tag{6}$$

According to the Bayesian formalism, the posterior $p(\mathbf{x}_i, \boldsymbol{\theta}_i \,|\, \mathbf{Y})$ may be expressed in terms of the likelihood function and the prior as

$$p(\mathbf{x}_i, \boldsymbol{\theta}_i \,|\, \mathbf{Y}) = \frac{p(\mathbf{Y} \,|\, \mathbf{x}_i, \boldsymbol{\theta}_i) p(\mathbf{x}_i, \boldsymbol{\theta}_i)}{p(\mathbf{Y})} \tag{7}$$

where $p(\mathbf{Y})$ is the marginal likelihood that results from integrating the likelihood over the hidden variables and the parameters.

3.2 Conjugate model

The likelihood function is jointly defined by the generative and the probabilistic models of the observable variables, the hidden variables and the model parameters. Given the noise in (4), the likelihood function for the AR1MA1 model is a multivariate Gaussian with unknown mean and variance as

$$p(\mathbf{Y} \,|\, \mathbf{x}_i, \boldsymbol{\theta}_i) = \mathrm{N}\left(\mathbf{Y} \,|\, \mathbf{R}\mathbf{D}_{\boldsymbol{\omega}_i}\mathbf{x}_i, \frac{\sigma_i^2}{\gamma_i} 1^N\right) \tag{8}$$

where $\mathbf{R} \in R^{N \times G}$ is a matrix with columns $[y_i(0), \ldots, y_i(N-1)]^\top$, $\mathbf{D}_{\boldsymbol{\omega}_i} = \mathrm{diag}(\boldsymbol{\omega}_i)$ is a square matrix with the elements of the vector $\boldsymbol{\omega}_i$ in its diagonal and γ_i a scale of the noise variance that depends on the variables and parameters as

$$\gamma_i \equiv \gamma_i(\mathbf{x}_i, \boldsymbol{\omega}_i) = \frac{1}{1 + \sum_{j=1}^{G} (x_i(j)\omega_i(j))^2}. \tag{9}$$

In order to favour an analytical solution, we use a conjugate model with Gaussian likelihood.

3.2.1 Conjugate priors

Although the hidden variables $\mathbf{x}_i$ are defined in a binary space, their priors are limited to a set of continuous distributions belonging to the conjugate exponential family. As is proposed in (Tienda-Luna *et al.*, 2007), we model the hidden variables by a Gaussian with mean $\mathbf{m}_{\mathbf{x}_i}$ and variance σ_0^2,

$$p(\mathbf{x}_i \,|\, \boldsymbol{\zeta}_{\mathbf{x}_i}) = \mathrm{N}(\mathbf{x}_i \,|\, \mathbf{m}_{\mathbf{x}_i}, \sigma_0^2 1^G) \tag{10}$$

with $\boldsymbol{\zeta}_{\mathbf{x}_i} = \{\mathbf{m}_{\mathbf{x}_i}, \sigma_0^2\}$ being the set of hyperparameters that model the hidden variables *a priori*.

On the other hand, the parameters $\boldsymbol{\theta}_i$ are modelled by a Normal scaled Inverse-Gamma distribution with mean $\mathbf{m}_{\boldsymbol{\omega}_i}$, variance scale γ_i, scale a_i and shape $\gamma_i b_i$ as

$$p(\boldsymbol{\theta}_i \,|\, \boldsymbol{\zeta}_{\boldsymbol{\theta}_i}) = p(\boldsymbol{\omega}_i \,|\, \sigma_i^2) p(\sigma_i^2) \tag{11}$$

$$= \mathrm{N}\left(\boldsymbol{\omega}_i \,|\, \mathbf{m}_{\boldsymbol{\omega}_i}, \frac{\sigma_i^2}{\gamma_i} 1^G\right) \mathrm{IG}(\sigma_i^2 \,|\, a_i, \gamma_i b_i)$$

with $\boldsymbol{\zeta}_{\boldsymbol{\theta}_i} = \{\mathbf{m}_{\boldsymbol{\omega}_i}, a_i, b_i\}$ being the set of prior hyperparameters of the model parameters.

3.2.2 Conjugate posteriors

According to the conjugate model proposed in (10) and (11), the posterior of the hidden variables and the parameters will belong to the same independent exponential families as their priors. Therefore, the posterior of the hidden variables is a Gaussian as

$$p(\mathbf{x}_i \,|\, \boldsymbol{\xi}_{\mathbf{x}_i}) = \mathrm{N}(\mathbf{x}_i \,|\, \boldsymbol{\mu}_{\mathbf{x}_i}, \Sigma_{\mathbf{x}_i}) \tag{12}$$

with $\boldsymbol{\xi}_{\mathbf{x}_i} = \{\boldsymbol{\mu}_{\mathbf{x}_i}, \Sigma_{\mathbf{x}_i}\}$ being the posterior hyperparameters of the hidden variables.

On the other hand, the posterior of the weights and the noise variance is a Normal scaled Inverse-Gamma as

$$p(\boldsymbol{\theta}_i \,|\, \boldsymbol{\xi}_{\boldsymbol{\theta}_i}) = p(\boldsymbol{\omega}_i \,|\, \sigma_i^2) p(\sigma_i^2) \tag{13}$$

$$= \mathcal{N}(\boldsymbol{\omega}_i \,|\, \boldsymbol{\mu}_{\boldsymbol{\omega}_i}, \sigma_i^2 \Sigma_{\boldsymbol{\omega}_i}) \mathrm{IG}(\sigma_i^2 \,|\, \alpha_i, \beta_i)$$

with $\boldsymbol{\xi}_{\boldsymbol{\theta}_i} = \{\boldsymbol{\mu}_{\boldsymbol{\omega}_i}, \Sigma_{\boldsymbol{\omega}_i}, \alpha_i, \beta_i\}$ being the posterior hyperparameters modelling the unknown parameters.

3.3 Variational Bayesian inference

The computation of the posterior from the likelihood and the priors is difficult in most general cases. The large number of variables considered in the GRN and the complex calculus of the marginal likelihood, to the point of being analytically intractable, demands the use of numerical approaches and computational methods. Alternatively to the computation of the posterior probability, the variational Bayesian Expectation-Maximization (VBEM) method optimizes a functional that depends on a free distribution of the hidden variables and the parameters as

$$F[q(\mathbf{x}_i, \boldsymbol{\theta}_i)] = \int d\mathbf{x}_i d\boldsymbol{\theta}_i q(\mathbf{x}_i, \boldsymbol{\theta}_i) \ln \frac{q(\mathbf{x}_i, \boldsymbol{\theta}_i)}{p(\mathbf{y}_i \,|\, \mathbf{x}_i, \boldsymbol{\theta}_i) p(\mathbf{x}_i, \boldsymbol{\theta}_i)}. \tag{14}$$

For any arbitrary free distribution $q(\mathbf{x}_i, \boldsymbol{\theta}_i)$, the functional in (14) is a lower bound of the logarithm of the marginal likelihood, which factorizes for each gene as

$$\ln\, p(\mathbf{y}_i) \geq F\,[q(\mathbf{x}_i, \boldsymbol{\theta}_i)] \tag{15}$$

where the optimum free distribution that maximizes (15) and holds the equality factorizes as

$$q(\mathbf{x}_i, \boldsymbol{\theta}_i) = p(\mathbf{x}_i \,|\, \boldsymbol{\xi}_{\mathbf{x}_i})\; p\;\left(\boldsymbol{\theta}_i \,|\, \boldsymbol{\xi}_{\boldsymbol{\theta}_i}\right). \tag{16}$$

The optimum free distribution that maximizes the variational lower bound can be computed by applying a set of sequential learning rules of the posterior hyperparameters $\boldsymbol{\xi}_{\mathbf{x}_i}$ and $\boldsymbol{\xi}_{\boldsymbol{\theta}_i}$. After each iteration of the VBEM method, the lower bound is updated and this procedure is repeated until it satisfies a given convergence criteria.

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For a detailed description about how to derive these learning rules and the lower bound update we refer to Tienda-Luna *et al.* (2007).

3.4 Approximated likelihood

The statistical model proposed earlier, with the scale of the variance γ_i defined as (9), breaks the symmetries of the conjugate exponential model. In order to support the conjugate modelling, we propose a fixed point approximation of the scale of the variance based on the MAP estimators of the hidden variables and the weights as

$$\gamma_i \equiv \gamma_i(\mathbf{x}_i, \boldsymbol{\omega}_i) \approx \gamma_i\left(\widehat{\mathbf{x}}_{i,\,\mathrm{MAP}}, \widehat{\omega}_{i,\,\mathrm{MAP}}\right) \tag{17}$$

where the MAP estimators can be approximated by the means $\widehat{\mathbf{x}}_{i,\,\mathrm{MAP}} \approx \boldsymbol{\mu}_{\mathbf{x}_i}$ and $\widehat{\omega}_{i,\,\mathrm{MAP}} \approx \boldsymbol{\mu}_{\boldsymbol{\omega}_i}$ learned at the latest iteration of the VBEM method. Therefore, the approximated scale of the variance can be computed after each VBEM interaction as

$$\gamma_i \approx \frac{1}{1 + \sum_{j=1}^{G}\left(\mu_{x_i(j)}\mu_{\omega_i(j)}\right)^2}. \tag{18}$$

4 Results

We have validated the AR1MA1-VBEM method by simulations and we have compared its performance with other approaches using in-silico data. In addition, we have analysed single cell qPCR data from mouse embryo during the development from oocyte to blastocyst (Guo *et al.*, 2010). Prior to the analysis using the AR1MA1-VBEM method, we applied tSNE (Maaten *et al.*, 2008) to reduce the data dimension and selected a subpopulation of cells differentiating to primitive endoderm (PE) cells. We then pseudo-temporally sorted the data by applying the modular optimal leaf-ordering algorithm (MOLO) (Sakai *et al.*, 2014). The AR1MA1-VBEM method for the GRN inference is implemented in MATLAB and is freely available at https://github.com/mscastillo/VBEM. Its low computational cost has allowed us to carry out all the analyses on a desktop computer.

4.1 Simulations

We have simulated gene expression time series for different scenarios, representing datasets with a variable number of genes and time samples. Given G genes, we have generated the topology of the network for a gene of interest, with the number of parents limited to $\frac{G}{3}$ to mimic the sparse structure of GRNs. The rest of the network settings, the regulatory type and the topology of rest of the genes, are generated using the priors in (10) and in (11). Once the genetic network is simulated, the synthetic dataset is generated according to the likelihood in (8), with N time samples and different levels of noise. Instead of working directly with the variance of noise, we have consider the signal-to-noise ratio (SNR) that represents the logarithm of the variance of the noise relative to the variance of the signal,

$$\mathrm{SNR} := -10\log_{10}\sigma_i^2. \tag{19}$$

Gene expression datasets have been simulated with SNR $= 1, \ldots, 40$. For each dataset, the topology of the network of the i'-th gene has been inferred using the AR1MA1-VBEM method, with a binary decision criteria $p(x_i(j) = 1) \geq \frac{1}{2}$. The results are analysed in the information retrieve context, where $x_i(j) = 1$ is considered as a positive (P) and $x_i(j) = 0$ is a negative (N). Therefore, two types of errors can be distinguished. A false positive (FP) consist of a relationship erroneously established between two genes as positive, when it is in fact negative whilst a false negative (FN) is a relationship erroneously established as negative, when actually is it positive. On the other hand, two types of hits are expected. The true positive (TP) and the true negative (TN), respectively, when a positive and a negative are correctly estimated.

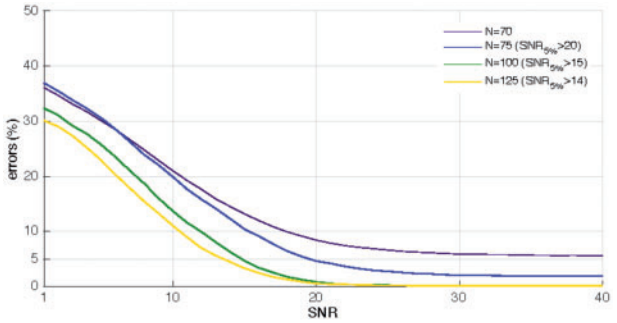


Fig. 1. Error in the GRN estimation with the AR1MA1-VBEM method, with a dataset with $G = 100$ genes and $N = \{125, 100, 75, 50\}$ time samples. $\mathrm{SNR}_{5\%}$ is the minimum SNR (if any) at which a satisfactory result is achieved under the 5% error level

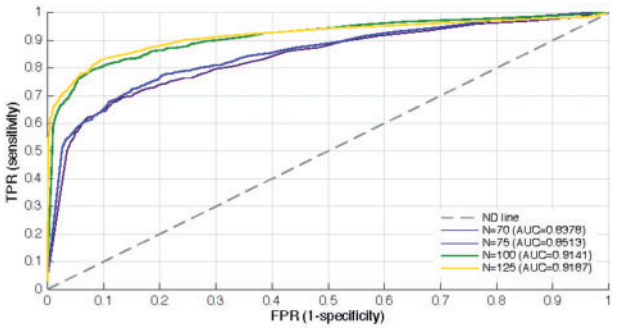


Fig. 2. Results from ROC analysis (FPR versus TPR) for the GRN estimation with the AR1MA1-VBEM method, with a dataset with $G = 100$ genes, SNR $= 10$ and $n = \{125, 100, 75, 50\}$ time samples

First, we analyse the whole number of errors, .e. the FN plus the FP, versus the SNR. The inference process is repeated 100 times and the number of errors is averaged for each scenario. In Figure 1 we plot the percentage of errors, over the 100 realizations, for a dataset with $G = 100$ genes and different number of time samples N versus the SNR. The 50% level corresponds to the limit of a random performance. On the contrary the optimum performance would have a 0% error level. Results show a dependency with the number of time samples and the level of noise. We consider a satisfactory result if the errors are under the 5% level at any SNR. It is clear that the AR1MA1-VBEM method provides satisfactory results with a rate $\frac{N}{G} \geq 0.75$.

Second, we use the receiver operating characteristic (ROC) curve to analyse the trade-off between the hits and the errors when the binary detection criteria changes. The ROC curve represents the true positive rate (TPR) versus the false positive rate (FPR) when the binary detection criteria changes. Intuitively, the ROC curve illustrates the specificity cost of the method, defined as 1–FPR, when its sensitivity is increased. The random performance is depicted in the ROC space by a line through the origin with unitary slope, known as the no-discrimination (ND) line, whilst the optimum performance would have a constant TPR = 1 at any FPR.

Figure 2 shows the ROC curve for scenarios with $G = 100$, SNR $= 10$ and different number of time samples N. ROC analysis summarizes the performance with the area under the ROC curve

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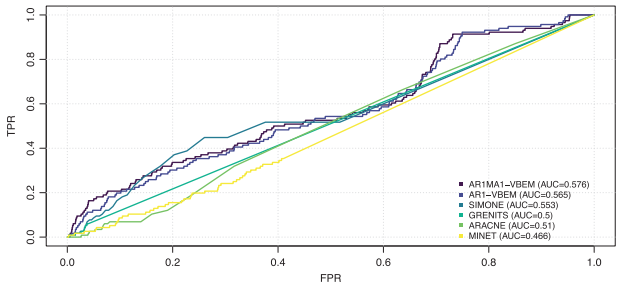


Fig. 3. Results from ROC analysis (FPR versus TPR) for the GRN inference comparison with $G = 100$ genes and $\frac{N}{G} = 0.75$ time samples

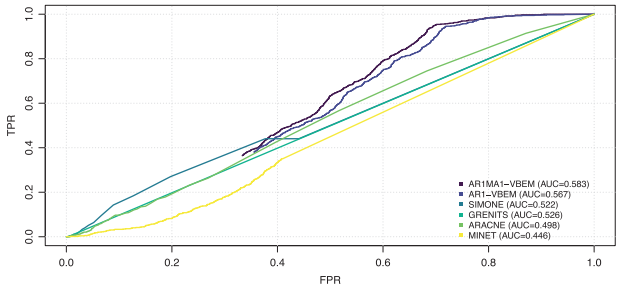


Fig. 4. Results from ROC analysis (FPR versus TPR) for the GRN inference comparison with $G = 500$ genes and $\frac{N}{G} = 0.75$ time samples

(AUC), which takes AUC $= \frac{1}{2}$ for the no discrimination (ND) line and a top value of AUC $= 1$ for the optimum performance. Our method shows a good performance that depends on the number of time samples.

4.2 *In silico* data

We have compared the performance of the proposed method with other approaches using synthetic datasets. Specifically, we have compared AR1MA1-VBEM with two other methods based on autoregressive models: the AR1-VBEM method presented in (Tienda-Luna *et al.*, 2007) and GRENITS which considers an AR1 model and a Gibbs sampler solution (Morrissey *et al.*, 2010). Additionally, we have considered other methods based on different approaches: SIMONE (Chiquet *et al.*, 2009) that combines the estimation of sparse undirected graphs with mixture model-based network clustering, ARACNE (Margolin *et al.*, 2006) and MINET (Meyer *et al.*, 2008) both based on mutual information measures between gene pairs to identify significant interactions.

Synthetic datasets were simulated with GeneNetWeaver (Schaffter *et al.*, 2011) using the transcriptional regulatory network of *Escherichia coli* and the multifactorial experiments (time series showing how the network responds to a random perturbation of a subset of genes by a small amount). Specifically, we have selected two sub-networks of different sizes with $G = 100$ and $G = 500$ genes. Data were generated using multifactorial time series experiments with default parameters, additive noise and with a lesser number of time samples than the number of genes, using the ratio $\frac{N}{G} = 0.75$. Fold changes were computed for each single gene by taking the $\log_2$ differences between the experimental and the wild-type samples. Figures 3 and 4 show the ROC curves for each method applied to both datasets.

ARACNE and MINET provide the weakest performance in both scenarios, to the point of being under the random performance limit. These results can be explained as both methods are based on static models, so such a performance could be expected. Amongst the dynamic methods considered in the comparisons, AR1MA1-VBEM is the one that presents better results, outperforming the results with SIMONE and our previous AR1-VBEM method.

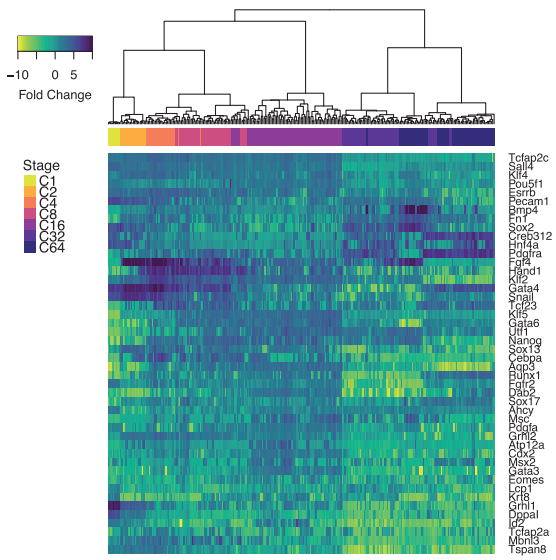


Fig. 5. Heatmap with $G = 46$ genes quantified in the mouse embryo experiment and $n = 280$ samples. Samples at columns are hierarchically clustered and pseudo-temporally sorted. Annotations at columns denote the known differentiation stages

4.3 *In vivo* data

We have tested the AR1MA1-VBEM method using HTSC data from mouse embryonic cells (Guo *et al.*, 2010) and hematopoetic cells in zebra fish (Macaulay *et al.*, 2016).

4.3.1 Single cell qPCR data in mouse embryo

Mouse embryonic data consists of 48 genes and 442 sample profiles during the 1- to 64-cell blastocyst transition (Guo *et al.*, 2010). Prior to the 32-cell stage, mouse embryo consists of two well differentiated cell types: the trophectoderm (TE) and the inner-cell mass (ICM). The ICM cells then differentiate into PE and epiblast (EPI) (Buettner *et al.*, 2012). Experimentally, samples can be differentiated by the expression of specific molecular markers that are specific to a cell type. For example, the transition from an ICM to PE or EPI is characterized by the expression of Gata4 and Nanog, respectively (Guo *et al.*, 2010). Despite single-cell data from a population not constituting an actual time series, this type of experiment can be sorted according to a pseudo-temporal index. Recent methods such as tSNE (Bendall *et al.*, 2014), Monocle (Trapnell *et al.*, 2014) and Diffusion Maps (Haghverdi *et al.*, 2015) allow the dissection and sorting of single-cell data. We refer to Cannoodt *et al.* (2016) for a review and comparison of pseudo-time ordering methods. Specifically, we have used tSNE to reduce the dimensionality of the dataset to two dimensions, prior data normalization as in (Guo *et al.*, 2010), and we have identified three sub-populations corresponding to cells prior to the 32-cell stage and cells differentiating to PE and EPI (Buettner *et al.*, 2012).

We have selected a dataset with $n = 280$ samples (comprising all the 1- to 16-cell and the 32- to 64-cell from the PE sub-population) and $G = 46$ genes (without taking into account the housekeeping genes). Figure 5 shows a heat map with this data, where samples are hierarchically clustered and sorted by the MOLO algorithm (Sakai

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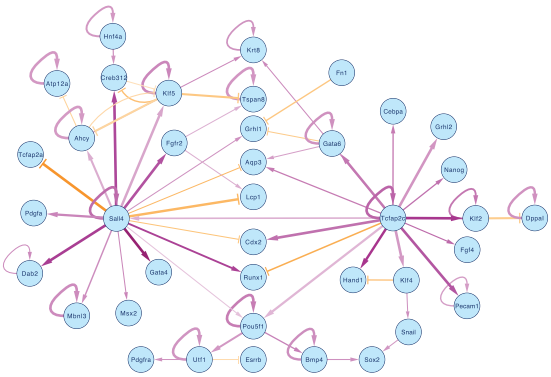


Fig. 6. GRN inferred by the AR1MA1 for mouse embryo. Activation relationships are represented as arrows and inhibition as *T*-shaped arrows. The width of the arrows is proportional to the posterior probability (the bigger, the more probable) and the colours are proportional to the weights (the darker, the bigger)

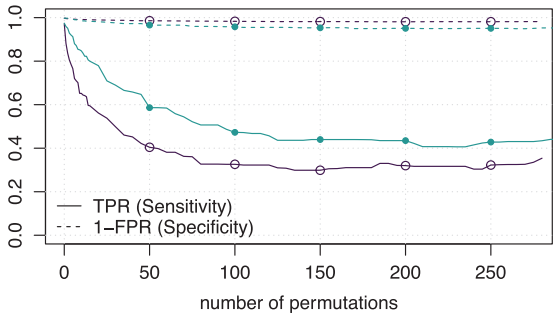


Fig. 7. Error rate of predictions of the GRN as the pseudo-time order changes. Results for mouse embryo (circles) and for Zebra fish (solid dots)

et al., 2014). MOLO provides similar results to the diffusion maps (Spearman's correlation $\rho = 0.71$, P-value $< 10^{-16}$) and successfully sorts data according to their differentiation stage.

We sorted data according to pseudo-time from MOLO as shown in Figure 5 and we ran the AR1MA1-VBEM method using non-informative priors. We then inferred the GRN by using a posterior probability threshold higher than 0.5. Figure 6 shows the inferred GRN comprising 37 genes and 66 relationships.

The inferred GRN captures the active role of both Tcfap2c and Sall4 during the development of mouse embryonic cells, represented as hub genes in the network. These two genes have well differentiated roles in ICM and TE differentiation as stated by their ontologies, being uniquely annotated in trophectodermal cell differentiation (GO: 0001829) or inner cell-mass cell proliferation (GO: 0001833), respectively. Our results show the up-regulation of Cdx2 by Tcfap2c and its down-regulation by Sall4. Cdx2 is a transcription factor from the AP-2 family which has been reported as a key factor for the maintenance of TE formation (Kuckenberg *et al.*, 2014). Sall4 up-regulates the expression of Gata4, a known PE-differentiation marker. This result suggests a potential mechanism by which Sall4 may repress TE differentiation and promotes cells to differentiate into PE. In addition, Sall4 down-regulates Tcfap2a (another transcription factor from the AP2-family) that potentially controls the transcriptional regulation of Cdx2.

We have analysed the dependency of the AR1MA1-VBEM method to variations of pseudo-time order. Specifically, we randomly permuted a number of paired samples from the MOLO ordering. We then inferred the GRN using the new ordered data and we compared it with the network in Figure 6. This process was repeated for a given number of permutations 100 times and the results analysed in terms of error rates. Figure 7 shows the sensitivity and specificity versus the number of permutations, from a single permutation to as many permutes as the number of samples. The results show that the AR1MA1-VBEM method has a strong dependency on the order to accurately infer the GRN, as shown by a drop in the sensitivity with the number of permutations. On the other hand, the AR1MA-VBEM method provides consistent results in terms of specificity, not inferring erroneous gene interactions as the pseudo-time order changes.

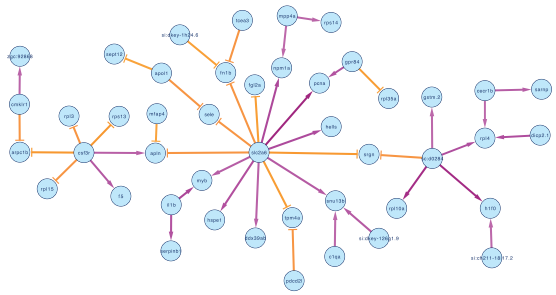


Fig. 8. GRN inferred by the AR1MA1 for zebra fish. Details as in Figure 6

4.3.2 Single cell RNA-Seq data in zebra fish

We have analysed data from zebra fish kidney cells pseudo-temporally sorted in Macaulay *et al.* (2016) using a mixed computational and experimental approach. Briefly, the dataset consists of $n = 363$ samples clustered in 4 groups associated with low and high expression of CD41. Subsequently, samples were pseudo-temporally sorted based on which cluster they belong to and the variability among components. The proposed order was validated with the pseudo-time ordering inferred with a Bayesian Gaussian process latent variable model, providing a Spearmanpe correlation $\rho = 0.97$. The final datasets consists of all the samples sorted according to the pseudo-time ordering proposed in Macaulay *et al.* (2016) and $G = 491$ unique genes identified as relevant cell-type markers for each of the clusters.

We ran the AR1MA1-VBEM method using non-informative priors and a probability threshold higher than 0.9, resulting in a dense network with more than 10 000 of interactions among all the genes. We have selected the top genetic relationships with highest posterior probability and highest weight that holds a fully connected module, leading to the GRN shown in Figure 8 with 45 genes and 44 edges. The module includes several ribosomal proteins (rpl3, rpl4, rps13, etc.) and slc2a6 as a hub, a gene involved in glucose transmembrane transport, which may facilitates rapid ribosome biogenesis. In addition, csf3r (colony stimulating factor 3 receptor) which controls the differentiation and function of granulocytes is also a hub. These genes may represent two key transcriptional mechanisms that correspond to a more stem and a more differentiated stages form the myeloid lineage.

We have analysed the dependency of the AR1MA1-VBEM method to variations of pseudo-time order as we did for the mouse embryo data using the results in Figure 8 as the reference network. Results in Figure 7 show similar results than for the mouse embryo data, with a strong dependency on the order to accurately infer the GRN and a robust performance in terms of specificity, not inferring erroneous gene interactions as the pseudo-time order changes.

5 Discussion

We review previous studies on the modelling of gene-expression time series data and the inference of GRNs. One of the main

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motivations to review these studies has been the availability of new experimental procedures, such as single cell experiments, which are able to generate data with a large number of samples. Even when these data do not constitute an actual time series, it is possible to sort them according to a pseudo-temporal index. We propose to model time and pseudo-time series data by a AR1 moving-average model to learn about the GRN within a variational Bayesian framework, namely the AR1MA1-VBEM method. We have compared the performance of the proposed method with previous approaches using synthetic data, where the AR1MA1-VBEM one showed the best results with moderated improvement in the ROC analysis using different datasets. The results using synthetic data reveal new opportunities to improve the performance of current methods.

We have also applied the AR1MA1-VBEM method to single cell qPCR data to infer the GRN for mouse embryo development and single cell RNA-Seq data for zebra fish. By combining hierarchical clustering with MOLO, we have been able to estimate the pseudo-time and we have applied the AR1MA1-VBEM method to infer the GRN. The results show that the AR1MA1-VBEM method successfully infers the active role of Tcfap2c and Sall4 during the ICM to primitive-endoderm transition. For single cell RNA-Seq data, we infer a GRN module that includes potential key genes in the mitochondrial biogenesis and differentiation of granulocytes. When we analysed the effects of the pseudo-time in the inference of the GRN, our results showed a robust performance of the AR1MA1-VBEM method in terms of specificity, which keeps the number of FNs to a minimum level.

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